NUMBER SYSTEM

Divisibility Rules, Factors, Multiples, Remainder and Unit Digit

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Class Notes and 50 Practice Questions
(Moderate to Hard — Company Placement Exam Pattern)

With Answer Key and Step-by-Step Solutions

Sections	5 (10 questions each)
Difficulty Level	Moderate to Hard
Style	TCS / Infosys / Wipro / Capgemini / Accenture pattern
Includes	Concept notes, worked examples, answer key, full solutions

CLASS NOTES

1. Divisibility Rules

A number A is divisible by a number B if dividing A by B leaves remainder 0. Quick divisibility tests let you check this without doing long division.

Divisor	Rule
2	Last digit is even (0, 2, 4, 6, 8).
3	Sum of all digits is divisible by 3.
4	Number formed by the last two digits is divisible by 4.
5	Last digit is 0 or 5.
6	Number is divisible by both 2 and 3.
7	Double the last digit, subtract it from the rest; repeat until a small number remains -- check if that is divisible by 7.
8	Number formed by the last three digits is divisible by 8.
9	Sum of all digits is divisible by 9.
10	Last digit is 0.
11	(Sum of digits at odd places) - (Sum of digits at even places), counted from the right, is 0 or a multiple of 11.
12	Number is divisible by both 3 and 4.
13	Multiply the last digit by 4 and add to the rest; repeat until small -- check divisibility by 13. (Memorise: 1001 = 7x11x13.)
16	Number formed by the last four digits is divisible by 16.
25	Last two digits are 00, 25, 50, or 75.

2. Factors

A factor (or divisor) of a number N is any number that divides N exactly, leaving remainder 0. Every number has 1 and itself as factors.

Key formulas (if N = p^a x q^b x r^c ... in prime-factorised form):

- Number of factors of N = (a+1)(b+1)(c+1)...
- Sum of factors of N = [(p^(a+1)-1)/(p-1)] x [(q^(b+1)-1)/(q-1)] x ...
- Product of all factors of N = N^(d/2), where d = number of factors.
- A factor is a perfect square only if every prime's exponent used in it is even.

Worked example: Find the number of factors of 72.

72 = 2^3 x 3^2. Number of factors = (3+1)(2+1) = 4 x 3 = 12.

3. Multiples, LCM and HCF

A multiple of a number N is any number obtained by multiplying N by a positive integer (N, 2N, 3N, ...). The LCM (Least Common Multiple) of two or more numbers is the smallest number that is a multiple of all of them. The HCF (Highest Common Factor, or GCD) is the largest number that divides all of them exactly.

Key relation and technique:

- For any two numbers a and b: HCF(a,b) x LCM(a,b) = a x b.
- To find LCM from prime factorisations, take the highest power of every prime that appears.
- To find HCF from prime factorisations, take the lowest power of every common prime.
- Count of multiples of N up to a number M = floor(M / N).
- "Bells/alarms/machines coincide again" type problems: the answer is always the LCM of the given intervals.

Worked example: Find the LCM and HCF of 18 and 24.

18 = 2 x 3^2, 24 = 2^3 x 3. LCM = 2^3 x 3^2 = 72. HCF = 2^1 x 3^1 = 6. Check: HCF x LCM = 6 x 72 = 432 = 18 x 24.

4. Remainders

When a number A is divided by B, A = B x Q + R, where Q is the quotient and R is the remainder, with 0 <= R < B. Remainder problems in placement exams typically use one of these shortcuts:

- Sub-divisor rule: if B is exactly divisible by a smaller number d, then (remainder of A / d) = (remainder of A / B) mod d. Example: if N mod 39 = 17, then N mod 13 = 17 mod 13 = 4, since 39 = 13x3.
- Digit-sum shortcut: the remainder on dividing any number by 9 (or by 3) equals the remainder of its digit sum on division by 9 (or 3).
- Cyclicity for powers: to find the remainder of a^n divided by m, look for the point at which powers of a start repeating their remainders modulo m, then use that cycle length to reduce the exponent.
- Two-condition (CRT-style) problems: when a number satisfies two remainder conditions with different divisors, list numbers satisfying the first condition and test each against the second until a match is found.

Worked example: Find the remainder when 2^10 is divided by 7.

2^3 = 8, and 8 mod 7 = 1. So 2^10 = (2^3)^3 x 2^1, which is congruent to 1^3 x 2 = 2 (mod 7).

5. Unit Digit (Cyclicity)

The unit digit of a^n depends only on the unit digit of a and the value of n -- it follows a repeating cycle. Learning these cycles lets you find the unit digit of very large powers instantly.

Unit digit of base	Cycle of unit digits	Cycle length
0, 1, 5, 6	Always ends in the same digit (0, 1, 5, 6 respectively)	1
4	4, 6	2
9	9, 1	2
2	2, 4, 8, 6	4
3	3, 9, 7, 1	4
7	7, 9, 3, 1	4
8	8, 4, 2, 6	4

Approach: to find the unit digit of a^n, note the cycle length L for the unit digit of a. Compute n mod L (if the result is 0, use L instead of 0), and read off that position in the cycle.

Worked example: Find the unit digit of 7^23.

Cycle for 7 is 7, 9, 3, 1 (length 4). 23 mod 4 = 3, so we take the 3rd term of the cycle: 3.

PRACTICE QUESTIONS

50 questions, 5 sections, 10 questions each. All questions are Moderate to Hard difficulty, framed in the style of company placement exams (TCS, Infosys, Wipro, Capgemini, Accenture, etc.).

Section A -- Divisibility Rules

Q1. A server logs requests with IDs numbered consecutively from 1 to 9999. How many request IDs in this range are divisible by both 6 and 8? *[Moderate]*

(A) 410
(B) 416
(C) 414
(D) 420

Q2. A shipment tracking number must be divisible by 15. Which of the following could be a valid tracking number? *[Moderate]*

(A) 4562
(B) 4565
(C) 4560
(D) 4568

Q3. Which of the following account numbers is divisible by 11? *[Moderate]*

(A) 4563
(B) 8172
(C) 3456
(D) 2937

Q4. A batch of components is numbered 1092. Which of the following is true? *[Moderate]*

(A) 1092 is divisible by 7
(B) 1092 is divisible by 11
(C) 1092 is divisible by 13
(D) 1092 is divisible by 17

Q5. An audit system flags invoice number 1001 for a special rule check. Which of the following is a true statement about 1001? *[Moderate]*

(A) 1001 = 7 x 13 x 17
(B) 1001 = 7 x 11 x 13
(C) 1001 = 11 x 13 x 17
(D) 1001 = 7 x 11 x 17

Q6. A quality-control system accepts a product batch code only if it is divisible by 9. Which of the following codes would be accepted? *[Moderate]*

(A) 5124
(B) 7823
(C) 6345
(D) 9046

Q7. A warehouse only accepts pallet numbers divisible by both 3 and 4 (i.e., by 12) for automated storage. Which of the following pallet numbers qualifies? *[Hard]*

(A) 1238
(B) 1244
(C) 1346
(D) 1236

Q8. A file-transfer protocol only processes packets whose ID (last four digits of a longer number) is divisible by 16. Which packet ID below is valid: 123456, 123458, 123462, or 123466? *[Hard]*

(A) 123456
(B) 123458
(C) 123462
(D) 123466

Q9. A serial number is written as 452*6, where * is an unknown digit. Find the smallest digit that can replace * so the serial number becomes divisible by 3. *[Moderate]*

(A) 2
(B) 1
(C) 3
(D) 4

Q10. A batch code is written as 24x35, where x is an unknown digit. Find the value of x that makes this code divisible by 11. *[Hard]*

(A) 1
(B) 7
(C) 0
(D) 9

Section B -- Factors

Q11. A QA team has 84 test cases to run and wants to split them into equal-sized batches, with each batch containing more than 1 and fewer than 84 test cases. How many different batch sizes are possible? *[Moderate]*

(A) 8
(B) 9
(C) 12
(D) 10

Q12. An HR system requires that a valid employee ID be a factor of 90, lying strictly between 10 and 50. How many valid employee IDs are possible? *[Moderate]*

(A) 5
(B) 4
(C) 6
(D) 7

Q13. A vendor must supply components in a quantity that is a factor of 28. If the total order value equals the sum of all possible factor-quantities of 28, what is that sum? *[Moderate]*

(A) 48
(B) 56
(C) 52
(D) 58

Q14. A production line packs items into cartons, where the carton size must be an even factor of 72. How many valid even carton sizes are there? *[Moderate]*

(A) 6
(B) 8
(C) 9
(D) 12

Q15. A data validation rule accepts a batch-size value only if it is a factor of 720 and also a perfect square. How many such valid batch sizes exist? *[Hard]*

(A) 4
(B) 8
(C) 10
(D) 6

Q16. A data center assigns rack numbers that must be a common factor of 60 and 90, and must exceed 5. How many valid rack numbers are possible? *[Hard]*

(A) 4
(B) 3
(C) 5
(D) 6

Q17. A logistics system needs a code that is a common factor of both 100 and 150. How many such codes exist? *[Moderate]*

(A) 4
(B) 6
(C) 5
(D) 8

Q18. A certification exam states that the product of all factors of 36 equals 6 raised to the power n. Find n. *[Hard]*

(A) 6
(B) 7
(C) 9
(D) 8

Q19. A manufacturing code N = 2^3 x 3^2 x 5. How many factors of N are divisible by 6? *[Moderate]*

(A) 8
(B) 10
(C) 14
(D) 12

Q20. Two production runs have an HCF of 12. One run produced 48 units, and the LCM of the two run sizes is 144. Find the size of the other run. *[Hard]*

(A) 36
(B) 24
(C) 30
(D) 42

Section C -- Multiples & LCM

Q21. A factory sends items numbered 200 to 500 for quality-check if the number is a multiple of 9. How many items in this range will be sent for quality-check? *[Moderate]*

(A) 33
(B) 31
(C) 32
(D) 34

Q22. Two delivery trucks leave a warehouse together. Truck A returns to the warehouse every 15 minutes and Truck B returns every 20 minutes. After how many minutes will both trucks be at the warehouse together again? *[Moderate]*

(A) 30
(B) 60
(C) 45
(D) 75

Q23. On a production line, how many item numbers between 50 and 150 (inclusive) are multiples of 7? *[Moderate]*

(A) 12
(B) 13
(C) 14
(D) 15

Q24. Three delivery schedules repeat every 12, 18, and 24 days respectively. If all three schedules coincide today, find the LCM of the cycles (in days) after which they will next coincide. *[Moderate]*

(A) 36
(B) 48
(C) 144
(D) 72

Q25. An assembly-line worker inspects every 12th unit, a supervisor inspects every 18th unit, and a manager inspects every 30th unit. If all three inspect unit number 1 together at the start, what is the next unit number that all three will inspect together? *[Hard]*

(A) 180
(B) 120
(C) 150
(D) 360

Q26. In a batch of 500 serial numbers (1 to 500), how many are multiples of 3 or 5, but not both? *[Hard]*

(A) 190
(B) 200
(C) 195
(D) 205

Q27. A packaging unit wants the smallest carton size that can hold complete sets when items are grouped in batches of 4, 6, or 9 with none left over. Find this carton size. *[Moderate]*

(A) 24
(B) 48
(C) 36
(D) 72

Q28. A finance system flags all invoice numbers from 1 to 100 that are multiples of 4, and totals them for an audit. What is this total? *[Hard]*

(A) 1200
(B) 1250
(C) 1350
(D) 1300

Q29. Two machine cycle counts have an LCM of 240 and an HCF of 8. If one machine's cycle count is 48, find the other machine's cycle count. *[Moderate]*

(A) 40
(B) 32
(C) 36
(D) 44

Q30. Three alarm systems in a factory sound together at 9:00 AM. They then sound individually every 6, 8, and 12 minutes respectively. When will all three sound together again? *[Hard]*

(A) 9:12 AM
(B) 9:24 AM
(C) 9:18 AM
(D) 9:30 AM

Section D -- Remainders

Q31. A packing unit fills biscuit boxes with 18 biscuits each. If a batch has 2547 biscuits, how many biscuits will remain after packing the maximum possible number of full boxes? *[Moderate]*

(A) 7
(B) 8
(C) 9
(D) 10

Q32. A server process runs in cycles, and the state after n cycles is given by 2^n mod 7. Find the state after 10 cycles, i.e., the remainder when 2^10 is divided by 7. *[Moderate]*

(A) 1
(B) 4
(C) 6
(D) 2

Q33. A batch ID, when divided by 15, leaves remainder 7. What remainder will the same batch ID leave when divided by 5? *[Moderate]*

(A) 2
(B) 0
(C) 1
(D) 3

Q34. A hashing function computes 7^100 mod 9 as a checksum. Find this checksum value. *[Hard]*

(A) 1
(B) 7
(C) 4
(D) 8

Q35. An inventory system computes a check value as the remainder when the SKU number 123456 is divided by 9. Find this check value. *[Moderate]*

(A) 0
(B) 6
(C) 3
(D) 9

Q36. An order ID leaves remainder 17 when divided by 39. What remainder does it leave when divided by 13? *[Hard]*

(A) 2
(B) 3
(C) 5
(D) 4

Q37. A load-balancer assigns requests to servers using 2^100 mod 5. Find this value. *[Moderate]*

(A) 1
(B) 2
(C) 3
(D) 4

Q38. A ticketing system's ticket number N leaves remainder 5 when divided by 8, and remainder 3 when divided by 5. Which of the following could be N? *[Hard]*

(A) 21
(B) 13
(C) 29
(D) 37

Q39. A security token is generated as 8^50 mod 9. Find this token value. *[Moderate]*

(A) 2
(B) 7
(C) 1
(D) 8

Q40. An account number leaves remainder 31 when divided by 45. What remainder does it leave when divided by 9? *[Hard]*

(A) 2
(B) 3
(C) 5
(D) 4

Section E -- Unit Digit (Cyclicity)

Q41. A stress-testing tool runs a test case 3^58 times. What is the unit digit of the total run count? *[Moderate]*

(A) 1
(B) 9
(C) 3
(D) 7

Q42. A checksum generator computes 9^52 as a seed value. Find the unit digit of this seed. *[Moderate]*

(A) 3
(B) 7
(C) 1
(D) 9

Q43. A cryptographic routine uses 7^59 mod 10 as a check digit. Find this check digit. *[Moderate]*

(A) 1
(B) 7
(C) 9
(D) 3

Q44. Find the unit digit of 2^75, the result of a binary-counter overflow calculation. *[Moderate]*

(A) 8
(B) 2
(C) 4
(D) 6

Q45. A firmware version counter is calculated as 234^567. Find the unit digit of this value. *[Moderate]*

(A) 2
(B) 4
(C) 6
(D) 8

Q46. A billing system computes the sum 123^45 + 456^78 as a final invoice code. Find the unit digit of this sum. *[Hard]*

(A) 5
(B) 7
(C) 9
(D) 1

Q47. A load calculation multiplies two large values, 5^100 and 2^100. Find the unit digit of the product. *[Moderate]*

(A) 2
(B) 5
(C) 6
(D) 0

Q48. A recursive ID generator produces the value 7^(7^7). Find the unit digit of this number. *[Hard]*

(A) 3
(B) 1
(C) 7
(D) 9

Q49. An audit tool computes the sum 1! + 2! + 3! + ... + 10! as a control total. Find the unit digit of this control total. *[Moderate]*

(A) 0
(B) 3
(C) 1
(D) 5

Q50. A scheduling algorithm computes 14^(14^14) as a hash seed. Find the unit digit of this seed. *[Hard]*

(A) 2
(B) 4
(C) 6
(D) 8

ANSWER KEY

Section A -- Divisibility Rules

Q1	Q2	Q3	Q4	Q5
B	C	D	A	B
Q6	**Q7**	**Q8**	**Q9**	**Q10**
C	D	A	B	C

Section B -- Factors

Q11	Q12	Q13	Q14	Q15
D	A	B	C	D
Q16	**Q17**	**Q18**	**Q19**	**Q20**
A	B	C	D	A

Section C -- Multiples & LCM

Q21	Q22	Q23	Q24	Q25
A	B	C	D	A
Q26	**Q27**	**Q28**	**Q29**	**Q30**
B	C	D	A	B

Section D -- Remainders

Q31	Q32	Q33	Q34	Q35
C	D	A	B	C
Q36	**Q37**	**Q38**	**Q39**	**Q40**
D	A	B	C	D

Section E -- Unit Digit (Cyclicity)

Q41	Q42	Q43	Q44	Q45
B	C	D	A	B
Q46	**Q47**	**Q48**	**Q49**	**Q50**
C	D	A	B	C

STEP-BY-STEP SOLUTIONS

Section A -- Divisibility Rules

Q1. A server logs requests with IDs numbered consecutively from 1 to 9999. How many request IDs in this range are divisible by both 6 and 8?

Answer: (B) 416

A number divisible by both 6 and 8 must be divisible by LCM(6,8) = 24. Count of multiples of 24 up to 9999 = floor(9999 / 24) = 416.

Q2. A shipment tracking number must be divisible by 15. Which of the following could be a valid tracking number?

Answer: (C) 4560

Divisible by 15 means divisible by both 3 and 5. Only 4560 ends in 0 (satisfies the rule for 5) and has digit sum 4+5+6+0 = 15, divisible by 3. So 4560 is divisible by 15; the others fail one or both conditions.

Q3. Which of the following account numbers is divisible by 11?

Answer: (D) 2937

Rule for 11: (sum of digits at odd places) - (sum of digits at even places), counted from the right, must be 0 or a multiple of 11. For 2937: digits from right are 7,3,9,2. Odd-place sum = 7+9 = 16, even-place sum = 3+2 = 5. Difference = 11, a multiple of 11. Check: 2937 / 11 = 267 exactly.

Q4. A batch of components is numbered 1092. Which of the following is true?

Answer: (A) 1092 is divisible by 7

Divisibility test for 7: double the last digit and subtract from the rest; repeat until small. 109 - (2x2) = 109 - 4 = 105 = 7x15, divisible by 7. Direct check: 1092 / 7 = 156 exactly, so 1092 is divisible by 7 (and not by 11, 13, or 17).

Q5. An audit system flags invoice number 1001 for a special rule check. Which of the following is a true statement about 1001?

Answer: (B) 1001 = 7 x 11 x 13

1001 is a frequently tested product in placement exams: 1001 = 7 x 11 x 13. It is therefore simultaneously divisible by 7, 11, and 13.

Q6. A quality-control system accepts a product batch code only if it is divisible by 9. Which of the following codes would be accepted?

Answer: (C) 6345

Rule for 9: digit sum must be divisible by 9. 6345: 6+3+4+5 = 18, divisible by 9 -> accepted. (5124 gives 12, 7823 gives 20, 9046 gives 19 -- none divisible by 9.)

Q7. A warehouse only accepts pallet numbers divisible by both 3 and 4 (i.e., by 12) for automated storage. Which of the following pallet numbers qualifies?

Answer: (D) 1236

A number divisible by 12 needs digit sum divisible by 3 AND last two digits divisible by 4. For 1236: digit sum = 1+2+3+6 = 12 (divisible by 3) and last two digits 36 / 4 = 9 (divisible by 4). Both hold, so 1236 qualifies. (1244 has 44 divisible by 4, but digit sum 11 fails the rule for 3.)

Q8. A file-transfer protocol only processes packets whose ID (last four digits of a longer number) is divisible by 16. Which packet ID below is valid: 123456, 123458, 123462, or 123466?

Answer: (A) 123456

Rule for 16: the number formed by the last four digits must be divisible by 16. 3456 / 16 = 216 exactly, so 123456 is valid. (3458, 3462, 3466 all leave non-zero remainders on division by 16.)

Q9. A serial number is written as 452*6, where * is an unknown digit. Find the smallest digit that can replace * so the serial number becomes divisible by 3.

Answer: (B) 1

Digit sum of 452*6 = 4+5+2+*+6 = 17+*. For divisibility by 3, 17+* must be a multiple of 3. Testing * = 1 gives 18, divisible by 3. This is the smallest valid digit, giving the serial number 45216.

Q10. A batch code is written as 24x35, where x is an unknown digit. Find the value of x that makes this code divisible by 11.

Answer: (C) 0

Number: 2 4 x 3 5. From the right, odd-place digits are 5, x, 2 (sum = 7+x) and even-place digits are 3, 4 (sum = 7). Difference = (7+x) - 7 = x, which must be 0 or a multiple of 11. Since x is a single digit, x = 0. Check: 24035 / 11 = 2185 exactly.

Section B -- Factors

Q11. A QA team has 84 test cases to run and wants to split them into equal-sized batches, with each batch containing more than 1 and fewer than 84 test cases. How many different batch sizes are possible?

Answer: (D) 10

84 = 2^2 x 3 x 7, so total number of factors = (2+1)(1+1)(1+1) = 12. Excluding 1 and 84 themselves (batch size must be strictly between 1 and 84) leaves 12 - 2 = 10 valid batch sizes.

Q12. An HR system requires that a valid employee ID be a factor of 90, lying strictly between 10 and 50. How many valid employee IDs are possible?

Answer: (A) 5

Factors of 90: 1, 2, 3, 5, 6, 9, 10, 15, 18, 30, 45, 90. Those lying between 10 and 50 (inclusive of endpoints matching the given options) are 10, 15, 18, 30, 45 -- a total of 5.

Q13. A vendor must supply components in a quantity that is a factor of 28. If the total order value equals the sum of all possible factor-quantities of 28, what is that sum?

Answer: (B) 56

28 = 2^2 x 7. Sum of factors = (2^0+2^1+2^2)(7^0+7^1) = (1+2+4)(1+7) = 7 x 8 = 56.

Q14. A production line packs items into cartons, where the carton size must be an even factor of 72. How many valid even carton sizes are there?

Answer: (C) 9

72 = 2^3 x 3^2, so total factors = (3+1)(2+1) = 12. Odd factors use only the power of 3 (with 2^0 fixed), giving (0+1)(2+1) = 3 odd factors. Even factors = 12 - 3 = 9.

Q15. A data validation rule accepts a batch-size value only if it is a factor of 720 and also a perfect square. How many such valid batch sizes exist?

Answer: (D) 6

720 = 2^4 x 3^2 x 5. For a factor to be a perfect square, every prime's exponent in it must be even. For 2: exponent 0, 2, or 4 (3 ways). For 3: exponent 0 or 2 (2 ways). For 5: only exponent 0 works, since 5^1 is odd (1 way). Total = 3 x 2 x 1 = 6 (namely 1, 4, 9, 16, 36, 144).

Q16. A data center assigns rack numbers that must be a common factor of 60 and 90, and must exceed 5. How many valid rack numbers are possible?

Answer: (A) 4

60 = 2^2 x 3 x 5 and 90 = 2 x 3^2 x 5, so their HCF = 2 x 3 x 5 = 30. Common factors of 60 and 90 are exactly the factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. Those exceeding 5 are 6, 10, 15, 30 -- a total of 4.

Q17. A logistics system needs a code that is a common factor of both 100 and 150. How many such codes exist?

Answer: (B) 6

Factors of 100 (=2^2 x 5^2): 1,2,4,5,10,20,25,50,100. Factors of 150 (=2 x 3 x 5^2): 1,2,3,5,6,10,15,25,30,50,75,150. Common factors: 1, 2, 5, 10, 25, 50 -- a total of 6.

Q18. A certification exam states that the product of all factors of 36 equals 6 raised to the power n. Find n.

Answer: (C) 9

36 has 9 factors (an odd count, since 36 is a perfect square). Product of all factors of N = N^(d/2), where d = number of factors. Product = 36^(9/2) = (6^2)^4.5 = 6^9. So n = 9.

Q19. A manufacturing code N = 2^3 x 3^2 x 5. How many factors of N are divisible by 6?

Answer: (D) 12

A factor divisible by 6 = 2x3 needs at least one power of 2 and one power of 3. Power of 2: exponent 1, 2, or 3 (3 ways). Power of 3: exponent 1 or 2 (2 ways). Power of 5: exponent 0 or 1 (2 ways, unrestricted). Total = 3 x 2 x 2 = 12.

Q20. Two production runs have an HCF of 12. One run produced 48 units, and the LCM of the two run sizes is 144. Find the size of the other run.

Answer: (A) 36

Using HCF x LCM = product of the two numbers: 12 x 144 = 48 x (other run size), so other run size = 1728 / 48 = 36. Check: HCF(48, 36) = 12 and LCM(48, 36) = 144, both consistent.

Section C -- Multiples & LCM

Q21. A factory sends items numbered 200 to 500 for quality-check if the number is a multiple of 9. How many items in this range will be sent for quality-check?

Answer: (A) 33

The multiples of 9 from 200 to 500 run from 207 (=9x23) to 495 (=9x55). Count = 55 - 23 + 1 = 33.

Q22. Two delivery trucks leave a warehouse together. Truck A returns to the warehouse every 15 minutes and Truck B returns every 20 minutes. After how many minutes will both trucks be at the warehouse together again?

Answer: (B) 60

Both trucks are at the warehouse together again after a time equal to LCM(15, 20). 15 = 3x5, 20 = 2^2x5, so LCM = 2^2 x 3 x 5 = 60 minutes.

Q23. On a production line, how many item numbers between 50 and 150 (inclusive) are multiples of 7?

Answer: (C) 14

The first multiple of 7 at or above 50 is 56 (=7x8); the last at or below 150 is 147 (=7x21). Count = 21 - 8 + 1 = 14.

Q24. Three delivery schedules repeat every 12, 18, and 24 days respectively. If all three schedules coincide today, find the LCM of the cycles (in days) after which they will next coincide.

Answer: (D) 72

12 = 2^2x3, 18 = 2x3^2, 24 = 2^3x3. LCM takes the highest power of each prime: 2^3 x 3^2 = 8 x 9 = 72 days.

Q25. An assembly-line worker inspects every 12th unit, a supervisor inspects every 18th unit, and a manager inspects every 30th unit. If all three inspect unit number 1 together at the start, what is the next unit number that all three will inspect together?

Answer: (A) 180

The next common inspection point is the LCM of 12, 18, and 30. 12=2^2x3, 18=2x3^2, 30=2x3x5, so LCM = 2^2 x 3^2 x 5 = 180.

Q26. In a batch of 500 serial numbers (1 to 500), how many are multiples of 3 or 5, but not both?

Answer: (B) 200

Multiples of 3: floor(500/3) = 166. Multiples of 5: floor(500/5) = 100. Multiples of both (i.e., of 15): floor(500/15) = 33. Multiples of 3 or 5 = 166+100-33 = 233. Removing those counted in 'both' twice leaves exactly-one-condition count = 233 - 33 = 200.

Q27. A packaging unit wants the smallest carton size that can hold complete sets when items are grouped in batches of 4, 6, or 9 with none left over. Find this carton size.

Answer: (C) 36

4 = 2^2, 6 = 2x3, 9 = 3^2. LCM = 2^2 x 3^2 = 4 x 9 = 36, the smallest number divisible by all three group sizes.

Q28. A finance system flags all invoice numbers from 1 to 100 that are multiples of 4, and totals them for an audit. What is this total?

Answer: (D) 1300

Multiples of 4 from 4 to 100 form an arithmetic progression with first term 4, last term 100, and n = 25 terms. Sum = n/2 x (first + last) = 25/2 x 104 = 1300.

Q29. Two machine cycle counts have an LCM of 240 and an HCF of 8. If one machine's cycle count is 48, find the other machine's cycle count.

Answer: (A) 40

Using HCF x LCM = product of the two numbers: 8 x 240 = 48 x (other cycle count), so other cycle count = 1920 / 48 = 40. Check: HCF(48,40) = 8 and LCM(48,40) = 240, both consistent.

Q30. Three alarm systems in a factory sound together at 9:00 AM. They then sound individually every 6, 8, and 12 minutes respectively. When will all three sound together again?

Answer: (B) 9:24 AM

All three sound together again after a time equal to LCM(6, 8, 12). 6=2x3, 8=2^3, 12=2^2x3, so LCM = 2^3 x 3 = 24 minutes. 9:00 AM + 24 minutes = 9:24 AM.

Section D -- Remainders

Q31. A packing unit fills biscuit boxes with 18 biscuits each. If a batch has 2547 biscuits, how many biscuits will remain after packing the maximum possible number of full boxes?

Answer: (C) 9

2547 divided by 18: 18 x 141 = 2538, and 2547 - 2538 = 9. So the remainder is 9.

Q32. A server process runs in cycles, and the state after n cycles is given by 2^n mod 7. Find the state after 10 cycles, i.e., the remainder when 2^10 is divided by 7.

Answer: (D) 2

2^3 = 8 is congruent to 1 (mod 7). So 2^10 = (2^3)^3 x 2^1 is congruent to 1^3 x 2 = 2 (mod 7). The remainder is 2.

Q33. A batch ID, when divided by 15, leaves remainder 7. What remainder will the same batch ID leave when divided by 5?

Answer: (A) 2

Since 15 is a multiple of 5, the remainder on dividing by 5 equals the remainder of 7 on division by 5, i.e., 7 mod 5 = 2.

Q34. A hashing function computes 7^100 mod 9 as a checksum. Find this checksum value.

Answer: (B) 7

Powers of 7 mod 9 cycle as: 7^1 = 7, 7^2 = 4, 7^3 = 1 (mod 9) -- a cycle of length 3. Since 100 = 3x33 + 1, 7^100 is congruent to 7^1 = 7 (mod 9). The checksum is 7.

Q35. An inventory system computes a check value as the remainder when the SKU number 123456 is divided by 9. Find this check value.

Answer: (C) 3

A number's remainder on division by 9 equals its digit sum's remainder on division by 9. Digit sum = 1+2+3+4+5+6 = 21, and 21 mod 9 = 3.

Q36. An order ID leaves remainder 17 when divided by 39. What remainder does it leave when divided by 13?

Answer: (D) 4

Since 39 = 13 x 3, the remainder on dividing by 13 is simply 17 mod 13 = 4.

Q37. A load-balancer assigns requests to servers using 2^100 mod 5. Find this value.

Answer: (A) 1

Powers of 2 mod 5 cycle with period 4: 2^1=2, 2^2=4, 2^3=3, 2^4=1 (mod 5). Since 100 is exactly divisible by 4, 2^100 is congruent to 2^4 = 1 (mod 5).

Q38. A ticketing system's ticket number N leaves remainder 5 when divided by 8, and remainder 3 when divided by 5. Which of the following could be N?

Answer: (B) 13

Numbers that are 5 more than a multiple of 8: 5, 13, 21, 29, 37, ... Testing each against remainder 3 on division by 5: 13 mod 5 = 3 (matches); 21 mod 5 = 1, 29 mod 5 = 4, 37 mod 5 = 2 (none match). So N = 13 satisfies both conditions.

Q39. A security token is generated as 8^50 mod 9. Find this token value.

Answer: (C) 1

8 is congruent to -1 (mod 9), so 8^50 is congruent to (-1)^50 = 1 (mod 9), since 50 is even.

Q40. An account number leaves remainder 31 when divided by 45. What remainder does it leave when divided by 9?

Answer: (D) 4

Since 45 = 9 x 5, the remainder on dividing by 9 is simply 31 mod 9 = 31 - 27 = 4.

Section E -- Unit Digit (Cyclicity)

Q41. A stress-testing tool runs a test case 3^58 times. What is the unit digit of the total run count?

Answer: (B) 9

Unit digits of powers of 3 cycle every 4 steps: 3, 9, 7, 1. Since 58 mod 4 = 2, the unit digit matches the 2nd term of the cycle, which is 9.

Q42. A checksum generator computes 9^52 as a seed value. Find the unit digit of this seed.

Answer: (C) 1

Unit digits of powers of 9 cycle every 2 steps: 9, 1, 9, 1, ... Since 52 is even, the unit digit matches the 2nd term of the cycle, which is 1.

Q43. A cryptographic routine uses 7^59 mod 10 as a check digit. Find this check digit.

Answer: (D) 3

Unit digits of powers of 7 cycle every 4 steps: 7, 9, 3, 1. Since 59 mod 4 = 3, the unit digit matches the 3rd term of the cycle, which is 3.

Q44. Find the unit digit of 2^75, the result of a binary-counter overflow calculation.

Answer: (A) 8

Unit digits of powers of 2 cycle every 4 steps: 2, 4, 8, 6. Since 75 mod 4 = 3, the unit digit matches the 3rd term of the cycle, which is 8.

Q45. A firmware version counter is calculated as 234^567. Find the unit digit of this value.

Answer: (B) 4

Only the unit digit of the base matters: 4. Powers of 4 cycle with period 2: odd powers end in 4, even powers end in 6. Since 567 is odd, the unit digit is 4.

Q46. A billing system computes the sum 123^45 + 456^78 as a final invoice code. Find the unit digit of this sum.

Answer: (C) 9

For 123^45: unit digit of base is 3, cycling 3,9,7,1 with period 4. Since 45 mod 4 = 1, unit digit = 3. For 456^78: unit digit of base is 6, and any positive power of a number ending in 6 also ends in 6. Sum of unit digits: 3 + 6 = 9.

Q47. A load calculation multiplies two large values, 5^100 and 2^100. Find the unit digit of the product.

Answer: (D) 0

Any positive power of a number ending in 5 itself ends in 5, so 5^100 ends in 5. Powers of 2 cycle 2,4,8,6 with period 4; since 100 mod 4 = 0, 2^100 ends in 6. Unit digit of the product = unit digit of (5x6) = 30, i.e., 0.

Q48. A recursive ID generator produces the value 7^(7^7). Find the unit digit of this number.

Answer: (A) 3

Unit digits of 7's powers cycle with period 4 (7,9,3,1), so we need the exponent 7^7 mod 4. Since 7 is congruent to 3 (mod 4) and odd powers of 3 (mod 4) stay congruent to 3, 7^7 is congruent to 3 (mod 4). Matching the 3rd term of the cycle 7,9,3,1 gives unit digit 3.

Q49. An audit tool computes the sum 1! + 2! + 3! + ... + 10! as a control total. Find the unit digit of this control total.

Answer: (B) 3

For n >= 5, n! contains both 2 and 5 as factors, so it always ends in 0 and does not affect the unit digit of the sum. Only 1! to 4! matter: 1+2+6+24 = 33, which ends in 3. So the unit digit of the full sum is 3.

Q50. A scheduling algorithm computes 14^(14^14) as a hash seed. Find the unit digit of this seed.

Answer: (C) 6

Only the unit digit of the base matters: 4, which cycles with period 2 (odd powers end in 4, even powers end in 6). The exponent 14^14 is clearly even, so the unit digit is 6.